SYBR Green qPCR

M11 | ver 1.3 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Measure target amplification using Hieff UNICON® ColorGPS qPCR SYBR Green Master Mix (No Rox) on a Bio-Rad CFX96. For gene-expression assays, use cDNA prepared in M10 and interpret results with the controls below.

Materials & Equipment

Item	Supplier / identifier	Use
Hieff UNICON® ColorGPS qPCR SYBR Green Master Mix (No Rox), 2×	Yeasen / 11188ES	qPCR premix; matched product [1]
Forward and reverse primers	Assay-specific; 10 µM stocks	One validated primer pair per reaction
cDNA, no-RT products and positive-control template	Sample-specific	Samples and assay controls
Sterilized Type II water	Laboratory-prepared	Reaction setup, template dilution and NTCs
Filter tips	Axygen	Aerosol-resistant tips compatible with the pipette volume
Optical PCR plate and seals	CFX96-compatible consumables	Optical sealing and amplification
CFX96, plate centrifuge and pipettes	Bio-Rad / laboratory equipment	Amplification, plate collection and setup

Before You Start

Use the 20 µL manufacturer recipe below as the standard scale. Select template volume and dilution from an assay dilution check; do not assume one template concentration suits all targets. [1]

For M10 products, 850 ng is the RNA input into the complete 20 µL RT reaction, not the cDNA mass loaded into each qPCR well. Track dilution and volume instead of assigning an unmeasured cDNA yield.

Thaw the premix, gently invert until homogeneous and briefly spin. Resolve visible precipitate before dispensing; avoid vortexing and strong light. Keep stocks at -25 to -15 °C. [1]

Plan independent biological samples and technical replicates separately. A practical plate layout uses triplicate wells per sample and primer pair; technical wells do not increase biological sample size. Include the controls on the next page.

Procedure

A. Assign samples and controls

1. Make a plate map before pipetting. Give every well a sample ID, target, dilution, replicate ID and control designation. Run each reference-gene assay in separate wells from its target-gene assay.

Control / group	How to prepare	What it establishes
NTC: every primer pair, every plate	Replace the complete template volume with sterilized Type II water. Keep primers, premix and total reaction volume unchanged.	Detects reagent/setup background, including contamination or primer dimers.
No-RT (NRT): RNA-derived assays	Use the matched M10 reaction made without RT enzyme, at the same dilution and volume as +RT. Test assays that could amplify DNA.	Assesses residual DNA contribution; water-only NTC cannot replace this control.
Positive amplification control	Use a previously verified target-containing template with the same primers. Handle after low-input samples.	Checks that the PCR assay works. A DNA control does not test the RT step.
Biological comparator / calibrator	Process the relevant untreated, vehicle or vector control through extraction, RT and qPCR alongside treated samples.	Defines the biological comparison; choose it from the experiment design.
Reference gene(s)	Measure validated stable transcript(s) in every biological sample.	Supports expression normalization; do not assume GAPDH or ACTB is stable.
Extraction blank, when processing contamination is a concern	Carry a sample-free extraction through RT and qPCR.	Locates contamination introduced before qPCR setup.
Standards / inter-run calibrator, when needed	Use a dilution series for efficiency or absolute quantification; repeat a common sample across plates for cross-run comparison.	Checks usable assay range or run-to-run consistency.

Control functions follow Bio-Rad guidance [2]. Include replicate control wells where plate capacity permits; flag amplification in a negative-control well rather than averaging it away.

B. Assemble the reaction

2. Prepare the selected template input. If dilution is needed, use sterilized Type II water and record the dilution factor. Use sterilized Type II water in place of template for NTC wells.

Component	Per 20 µL	Final
2× ColorGPS Master Mix (No Rox)	10 µL	1×
Forward primer, 10 µM	0.4 µL	0.2 µM
Reverse primer, 10 µM	0.4 µL	0.2 µM
Prepared DNA/cDNA or control	x µL	Assay-selected
Sterilized Type II water	(9.2 - x) µL	To volume
Total	20 µL	

For undiluted RT product, x must be ≤2 µL. If using diluted product, track the undiluted RT equivalent and validate the dilution. The same x applies to matched samples and controls. [1]

3. Prepare one shared mix per primer pair containing premix, primers and water, with sufficient overage for dispensing loss. Dispense (20 - x) µL, then add x µL of the assigned template or control with a fresh tip.

4. Seal the plate with a compatible optical seal, mix without splashing and briefly centrifuge to collect liquid and remove bubbles. Verify plate orientation, fill pattern and seal contact before loading.

C. Configure the CFX96 and run

5. Create a 20 µL run. Select SYBR detection (SYBR/FAM scan mode where available); do not add ROX or enable ROX normalization. Enter target/sample names and designate NTC, NRT, positive controls and standards in the plate setup. Set plate type to match the consumable.

Stage	Temperature	Time / repeats	Read
Initial denaturation	95°C	2 min; once	No
Denaturation	95°C	10 sec; 40 cycles	No
Annealing / extension	60°C	30 sec; 40 cycles	Yes, each cycle
Melt curve	Instrument default melt program	After amplification	At each melt step

Use the Yeasen cycling conditions above [1]. In a GOTO-style editor, repeat the two cycling steps 39 additional times for 40 total cycles. Retain the CFX software's standard melt program, inspect its range and increments, and save these settings with the run.

6. Confirm fluorescence acquisition at the 60°C step and during melting, check the plate map against the physical plate, and start the run. Save both the protocol and plate definition.

D. Review and export

7. Inspect amplification and melt curves together with NTC, no-RT and positive controls before accepting Cq values. Check technical-replicate agreement and use consistent baseline/threshold rules within each assay. Investigate discrepant wells; retain the reason for any exclusion.

8. Export raw run data, Cq values, amplification and melt curves, plate map and dilution records. For relative expression, normalize to validated reference genes and the planned biological comparator; use comparative Cq calculations only after confirming compatible amplification efficiencies. [2]

p.s.

Prepare NTC wells before handling template stocks. Use dedicated clean-area pipettes and fresh aerosol-resistant filter tips for every transfer; keep amplified material away from setup.

Keep no-RT and NTC controls distinct. An NTC tests qPCR setup background; a no-RT control tests whether the RNA preparation can generate signal without reverse transcription.

Do not open amplified plates in the pre-PCR area. Keep seals intact when archiving or disposing of them to limit carryover into later runs.

Expected Outcome

Obtain interpretable amplification curves and an assay-consistent melt peak in valid positive samples, with appropriate negative controls and reproducible technical wells. A single melt peak supports specificity but does not prove amplicon identity. Retain control outcomes with the reported expression or quantity results.

Troubleshooting

Observation	Check first	Action
NTC amplifies with a sample-like melt peak	Contaminated water, primers, premix, pipettes or aerosol carryover; plate-map errors.	Repeat with fresh aliquots and a clean setup area. Switching to aerosol-resistant filter tips may improve the result by reducing carryover [1]; it cannot remove contamination already in a reagent. Replace suspect reagents and rerun NTCs.
Late NTC signal with a lower or additional melt peak	Primer dimers versus other nonspecific products.	Optimize annealing conditions and primer concentration within 0.1-1 µM [1], or redesign primers. Filter tips do not correct primer dimers. Do not use a universal late-Cq cutoff to dismiss the signal.
No-RT control amplifies	Residual DNA, DNA-removal setup and primer specificity.	Review M10 Eraser treatment and RNA handling. Consider exon-junction primers when applicable; resolve the DNA contribution before interpreting transcript changes.
No signal, including positive control	Missing components, wrong detection channel, absent plate-read step or program error.	Check the saved setup and reagent identity. Repeat with a verified control after correcting the fault.
Sample fails but positive control works	Template dilution, inhibitors, RNA integrity and target abundance.	Compare matched template dilutions and reference-gene results. Review extraction and RT records before increasing input.
Replicates disagree or edge wells differ	Bubbles, incomplete mixing, pipetting, seal integrity and evaporation.	Check well images/curves and plate notes; repeat affected samples with fresh sealing and consistent dispensing.
Multiple melt peaks	Nonspecific products, dimers or DNA carryover.	Compare NTC/no-RT and positive controls; optimize or redesign the assay before quantification. Verify product identity if necessary.

Reference

[1] Yeasen. Hieff UNICON ColorGPS qPCR Master Mix (No Rox), 11188ES. IFU, internal version Ver.EN20260116, pp. 1-3. Accessed 2026-09-27. Open source

[2] Bio-Rad. qPCR Amplification. Assay setup, assay controls, reference genes and ROX requirements. Accessed 2026-09-27. Open source